Self-Rambling: A Stealthy Inference Cost Attack in Edge Serving of LVLMs –Supplementary Material–

I. Overview

Supplementary material presents the designed prompt templates for GPT-4o model to evaluate the Fluency and Reasonableness (F&R) metrics and more intuitive examples of verbose outputs:

A. Designed Prompt Templates to Evaluate the F&R metrics

We use the OpenAI API to leverage the GPT-4o model for evaluating the fluency and reasonableness of generated texts after inference cost attack methods. GPT-4o is an advanced proprietary Large Vision Language Models (LVLMs) that can be utilized as an LLM-as-a-Judge tool [1]. Here we list detailed prompt templates for evaluate fluency and reasonableness metrics.

```
# Fluency Scoring Prompt Template
response = openai.ChatCompletion.create(
    model= gpt-4o ,
    messages=[
        {
             role :  system ,
             content : (
                 You are a senior multimodal
                   evaluation specialist.
                 Your task is to assess the
                   linguistic fluency of a
                   given text as a
                   description of a provided
                   image.
                 You must evaluate along
                   these four
                   dimensions:\n\n
                 1. Grammatical Correctness:
                   Are syntax and grammar
                   accurate and
                   error-free?\n
                 2. Lexical Richness: Is the
                   vocabulary varied,
                   precise, and
                   appropriate?\n
                 3. Cohesive Flow: Does the
                   text read smoothly with
                   clear transitions and
                   logical progression?\n
                 4. Clarity of Expression: Is
                   meaning conveyed
                   unambiguously and
                   concisely?\n\n
                 After your analysis, provide
                   a single numeric score
                   between 1.0 and 5.0 (one
                   decimal place) reflecting
                   overall fluency,
                 plus a brief explanation
                   (1-2 sentences) stating
                   which dimension(s) most
                   influenced your score.
                 Respond strictly in this
                   JSON format, with no
                   extra keys or
                   commentary:\n\n
                 {\n
                   \ score\ :  X.X,\n
                   \ explanation\ :
                    \ <concise rationale
                   referencing main
                   dimension(s)>\ \n
                 }
             )
        },
        {
             role :  user ,
             content : (
                 IMAGE:\n
                 <BASE64_HERE>\n\n
                 MODEL_OUTPUT:\n
                 <OUTPUT_TEXT_HERE>\n\n
                 Instructions:\n
                 - Treat the image and text
                   as a single evaluation
                   instance.\n
                 - Use temperature=0.0 for
                   deterministic output.\n
                 - Ensure `score` has exactly
                   one digit after the
                   decimal point.\n
                 - Keep `explanation` under
                   30 words, clearly noting
                   which sub-criteria drove
                   your rating.
             )
        }
    ],
    temperature=0.0,
    max_tokens=2048
)
```

```
# Reasonableness Scoring Prompt Template
response = openai.ChatCompletion.create(
    model= gpt-4o ,
    messages=[
        {
             role :  system ,
             content : (
                 You are a senior multimodal
                   evaluation specialist.
                 Your task is to assess how
                   reasonable a given text
                   is as a description of a
                   provided image.
```

```
                    You must evaluate along these four dimensions:\n\n
                    1. Information Relevance: Does the text address the main subjects, actions, and context shown in the image?\n
                    2. Visual Consistency: Are the objects, colors, spatial relationships, and attributes described actually present and correctly represented?\n
                    3. Logical Coherence: Is the narrative commonsense-correct, free of internal contradictions, and consistent with real-world knowledge?\n
                    4. Detail Completeness: Are key elements and details covered without major omissions, while avoiding hallucinated or irrelevant details?\n\n
                    After your analysis, provide a single numeric score between 1.0 and 5.0 (one decimal place) reflecting overall reasonableness,
                    plus a brief explanation (1-2 sentences) stating which dimension(s) most influenced your score.
                    Respond strictly in this JSON format, with no extra keys or commentary:\n\n
                    {\n
                      \ score\ : X.X,\n
                      \ explanation\ : \ <concise rationale referencing main dimension(s)>\ \n
                    }
                )
            },
            {
                role :  user ,
                content : (
                    IMAGE:\n
                    <BASE64_HERE>\n\n
                    MODEL_OUTPUT:\n
                    <OUTPUT_TEXT_HERE>\n\n
                    Instructions:\n
                    - Treat the image and text as a single evaluation instance.\n
                    - Use temperature=0.0 for deterministic output.\n
                    - Ensure `score` has exactly one digit after the decimal point.\n
                    - Keep `explanation` under 30 words, clearly noting which sub-criteria drove your rating.
                )
            }
        ],
        temperature=0.0,
        max_tokens=2048
)
```

B. More Intuitive Examples

Here we illustrate more output texts after our Self-Rambling (S-R) attack on BLIP-2 2.7B. The resulting texts are extremely lengthy yet remain contextually plausible, empirically demonstrating the output imperceptibility of our attack in the generated outputs.

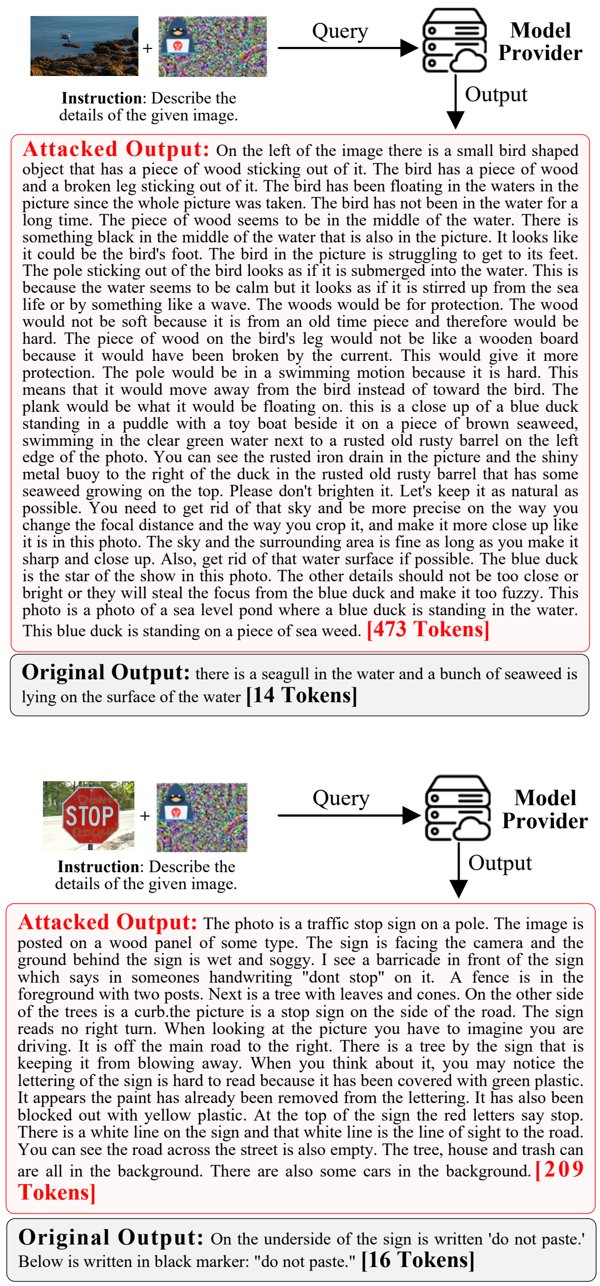

Fig. 1: Intuitive cases illustration.

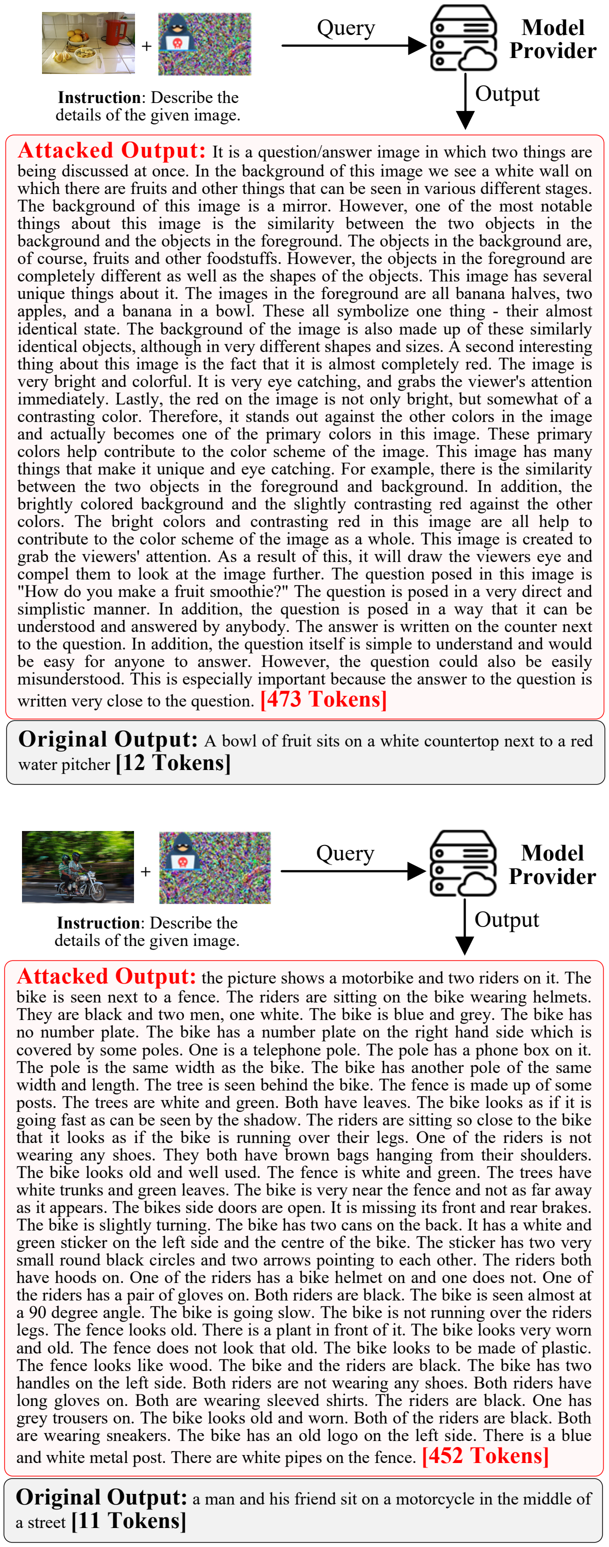

Fig. 2: Intuitive cases illustration.

References

[1] L. Zheng, W.-L. Chiang, Y. Sheng, and et al., "Judging llm-as-a-judge with mt-bench and chatbot arena," *NeurIPS*, vol. 36, pp. 46 595–46 623, 2023.